



IPRC KIGALI

Integrated Polytechnic Regional College

Email: info@iprckigali.rp.ac.rw

www.iprckigali.rp.ac.rw

Program: year 4 Btech in Electronics and Telecommunication Technology

Department: Electrical and Electronics Engineering

Option/Trade: Electronics and Telecommunication Technology

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Credits number: 10

Academic year: 2026/2027

Semester: I

Names: MUGISHA EMMANUEL

Reg no: 26RP00370

1st October 2026

MACHINE LEARNING ASSIGNMENT PROJECT REPORT

1. INTRODUCTION AND OBJECTIVES

This project develops a Rwanda house-price prediction application using multiple linear regression. The objective is to Estimate a house's market price in million Rwandan Francs from property characteristics. The workflow covers data Preparation, regression modelling, evaluation, model saving, GitHub version control and Streamlet deployment. The assignment requires the use of Area, Bedrooms, Bathrooms, House Age, and Distance to City, Parking Spaces and Neighborhood as predictors, while House ID is excluded from modelling.

2. DATASET AND DATA PREPARATION

The target variable is House_Price_Million_RWF. The predictors are Area_m2, Bedrooms, Bathrooms, House_Age_Years, Distance_to_City_km, Parking Spaces and Neighborhood. Neighborhood contains: Gasabo, Huye, Kicukiro, Kigali City, Musanze and Nyarugenge. Data cleaning included inspection of missing values, duplicate records, impossible or extreme observations, and categorical data. The cleaned dataset is stored in the project repository. The assignment requires four key visualizations: price distribution, Area versus Price, Price by Neighborhood and a correlation heat map.

Cleaning decision

Missing values should be handled according to the notebook analysis; duplicate records should be identified and removed where confirmed; extreme or impossible observations should be assessed using the chosen outlier method; Neighborhood should be one-hot encoded with one category dropped; and House ID should remain excluded from the model.

3. MODEL DEVELOPMENT

The model uses multiple linear regression. Neighborhood is converted to numerical dummy variables while avoiding the Dummy-variable trap. The data are divided into 80% training and 20% testing using random state=42. A scikit-learn Preprocessing pipeline is used so the saved model can accept raw input values directly.

4. MODEL EVALUATION AND INTERPRETATION

The required evaluation metrics are R^2 , adjusted R^2 , MAE and RMSE for both training and test sets. The assignment also requires an Actual-versus-Predicted plot, residual plots and interpretation of at least three coefficients.

Metric	Training	Testing
R^2	0.2830	-0.0474
Adjusted R^2	0.1794	-1.1901
MAE	37.5127	91.9796
RMSE	80.0294	218.7116

5. COEFFICIENTS AND DIAGNOSTICS

The final report should include the intercept and every fitted coefficient, followed by plain-language interpretation of at least three coefficients. Diagnostic analysis should discuss linearity, independence, constant variance, residual normality and multicollinearity. The required diagnostic plots should be copied from the notebook.

6. MODEL SAVING AND STREAMLIT APPLICATION

The trained pipeline is saved as **house_price_model.sav**. The Streamlit application loads the model with joblib and uses `@st.cache_resource`. It provides seven input widgets and displays the predicted price in million RWF.

7. GITHUB AND ONLINE DEPLOYMENT

The house price prediction project was maintained in a public GitHub repository to organize and manage the source code, trained machine learning model, dataset, Jupyter notebook, and supporting files.

The GitHub repository contains the following required files:

File	Purpose
app.py	Contains the Streamlit application code used to provide the house price prediction interface.
house_price_model.sav	Contains the trained machine learning model used to predict house prices.
requirements.txt	Contains the Python libraries and dependencies required to run the application.
cleaned_dataset.csv	Contains the cleaned dataset used for the machine learning project.
house_price_prediction.ipynb	Jupyter Notebook containing the data preparation, analysis, model training, and evaluation process.
README.md	Provides information about the project, installation requirements, usage instructions, and deployment details.

GitHub Repository

The project files were uploaded to a public GitHub repository, allowing the source code and supporting files to be accessed and reviewed online.

The development history was documented using at least four meaningful Git commits. The commits were used to record important stages of the project, such as adding the dataset and notebook, developing the prediction model, creating the Streamlit application, and preparing the project for deployment.

Online Deployment

The completed application was deployed online using Streamlit, allowing users to access the house price prediction system through a web browser without running the application locally.

The deployed application provides an interface where users can enter the required house characteristics and click the Predict House Price button to obtain a predicted house price.

GitHub Repository

<https://github.com/mugishalucky45-spec/house-price-predictor>

Streamlit Application

<https://house-price-predictor-erevqgvhqc29n7gmr2wrvl.streamlit.app/>

The screenshot displays the GitHub repository page for 'house-price-predictor' by user 'mugishalucky45-spec'. The repository is public and has 7 commits. The file list includes README.md, app.py, house_prediction.ipynb, house_price_model.sav, house_price_prediction_cleaned.csv, and requirements.txt. The README section is titled 'Rwanda House Price Predictor' and describes the project purpose, dataset, and features. The 'Suggested workflows' section on the right lists several workflows for configuration, including 'Python application', 'Django', and 'Python Package using Anaconda'.

Repository Overview:

- Repository: house-price-predictor (Public)
- Owner: mugishalucky45-spec
- Commits: 7
- Files: 6

File List:

File	Commit	Time
README.md	c8e0df3	4 hours ago
app.py	c8e0df3	3 hours ago
house_prediction.ipynb	c8e0df3	5 hours ago
house_price_model.sav	c8e0df3	4 hours ago
house_price_prediction_cleaned.csv	c8e0df3	5 hours ago
requirements.txt	c8e0df3	4 hours ago

README Content:

Rwanda House Price Predictor

Project Purpose

This project develops a machine learning model that predicts house prices in Rwanda using multiple linear regression. The prediction is based on house characteristics such as area, bedrooms, bathrooms, house age, distance to the city centre, parking spaces, and neighbourhood.

The trained model is also provided through an interactive Streamlit web application.

Dataset

The dataset contains house information and prices in million Rwandan Francs (RWF).

Features

- Area_m2
- Bedrooms
- Bathrooms
- House_Age_Years
- Distance_to_City_km
- Parking_Spaces
- Neighborhood

Target

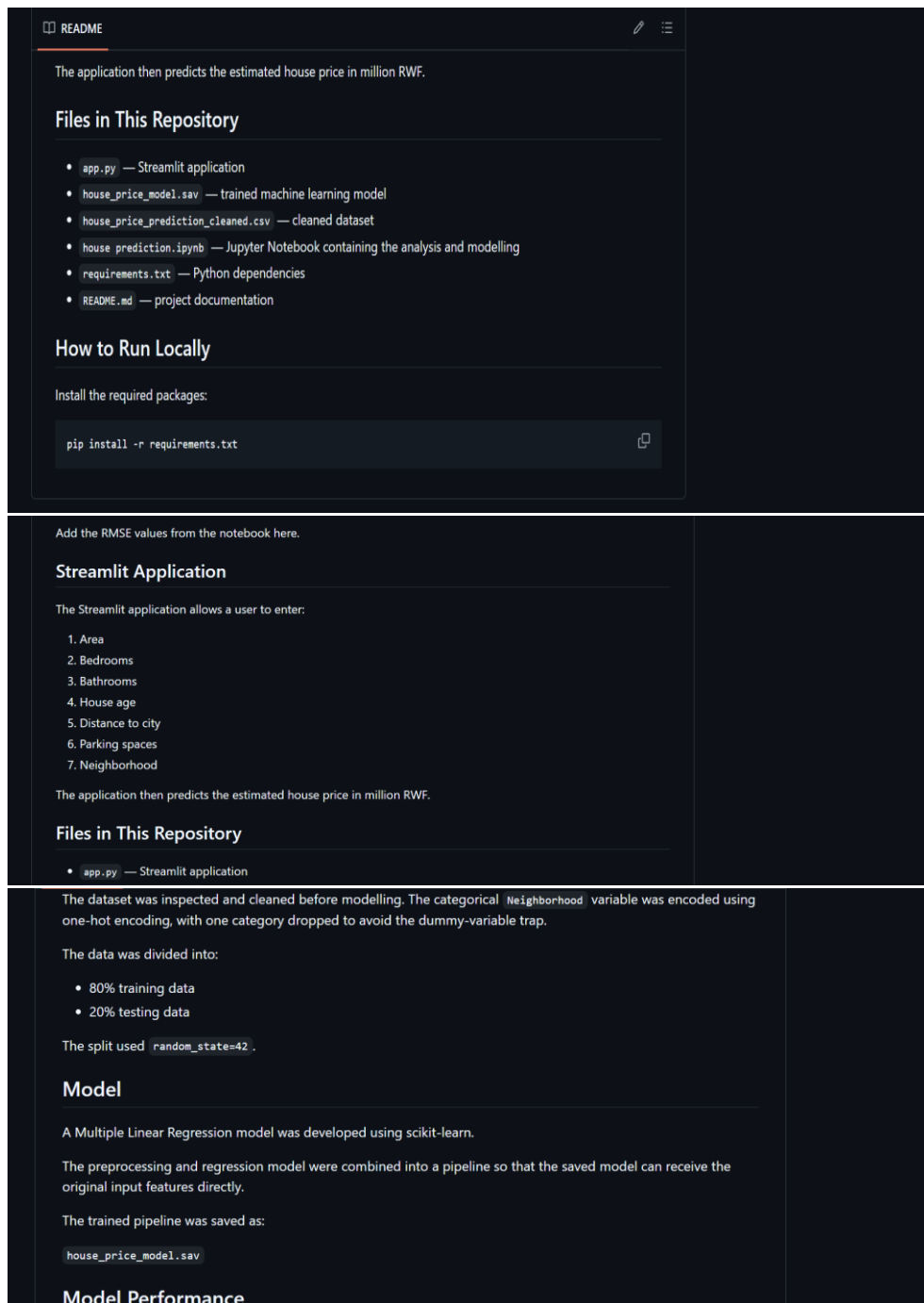
- House_Price_Million_RWF

House_ID was excluded from the model because it is only an identifier.

Data Preparation

Suggested workflows:

- Python application: Create and test a Python application. By GitHub Actions. [Configure](#)
- Django: Build and Test a Django Project. By GitHub Actions. [Configure](#)
- Python Package using Anaconda: Create and test a Python package on multiple Python versions using Anaconda for package management. By GitHub Actions. [Configure](#)



8. APPLICATION TESTING

The deployed house price prediction application was tested using valid combinations of house characteristics to verify that the system operates correctly and produces a house price prediction.

For the testing process, the user entered valid values for the required house characteristics in the online application and clicked the **“Predict House Price”** button. The application processed the input values using the trained machine learning model and displayed the predicted house price.

Test Case

Parameter	Test value
Area (sq ft)	190.00
Bedrooms	6
Bathrooms	5
House age(years)	9.00
Distance to city(km)	1.00
Parking spaces	3
Neighborhood	Nyarugenge
Prediction	291.65 million RWF
Result	Test passed

The application successfully accepted the valid input values and generated a house price prediction. This confirms that the deployed application is connected to the trained model and can provide predictions based on user-provided house characteristics.

Screenshot of Application Testing

A screenshot of the deployed online application showing the entered house characteristics and the resulting predicted house price should be inserted below.

Share

Rwanda House Price Predictor

Student Information

Student Name: MUGISHA EMMANUEL
Registration Number: 26RP00370
Module: MACHINE LEARNING
Module Code: GENML801
Department: ELECTRICAL AND ELECTRONIC DEPARTMENT
Option and Class: ETT Y4 BTECH
Date: 01ST OCTOBER 2026

Estimate the market price of a house in million RWF using a trained multiple linear regression model.

Area (m²)

190.00- +

Bedrooms

Bathrooms

5- +

House Age (years)

9.00- +

Distance to City (km)

1.00- +

Parking Spaces

3- +

Neighborhood

Nyarugenge

Predict House Price

Distance to City (km)

1.00- +

Parking Spaces

3- +

Neighborhood

Nyarugenge

Predict House Price

Estimated house price: 291.65 million RWF

9. CONCLUSION

This project successfully demonstrates an end-to-end machine-learning workflow for house price prediction in Rwanda. It covers the major stages of the machine-learning process, including data cleaning, exploratory data analysis, handling outliers, categorical encoding, model development using Multiple Linear Regression, and model evaluation using R^2 , Adjusted R^2 , MAE, and

RMSE. The project also incorporates model persistence, GitHub version control, and Streamlit deployment, demonstrating how a machine-learning model can be developed and made accessible through a user-friendly application.

The model uses relevant house characteristics, such as area, number of bedrooms and bathrooms, house age, distance to the city, parking spaces, and neighborhood, to estimate house prices. However, the evaluation results indicate that the model's predictive performance on the test dataset is limited. This suggests that the current model may not fully capture the factors influencing house prices in Rwanda. Further improvements could include collecting more representative data, investigating extreme house prices, selecting more relevant features, and comparing Multiple Linear Regression with other regression algorithms.

In conclusion, the project provides a practical foundation for applying machine learning to real estate price estimation in Rwanda. Although the current model requires further improvement before it can be relied upon for accurate price estimates, the developed workflow demonstrates the potential of data-driven techniques to support property valuation and provides a basis for future research and development.

10. Limitations and Future Improvements

The developed house price prediction model has several limitations. First, the dataset contains a limited number of housing records, which may not fully represent the diversity of the real estate market in Rwanda. Second, the model uses a limited set of house characteristics and may not capture other important factors affecting property prices, such as property condition, exact location, land size, finishing quality, access to essential services, and changes in market conditions. In addition, the current Multiple Linear Regression model has limited predictive performance on the test dataset, indicating that further improvements are necessary before it can be used for reliable price estimation.

Future improvements could include collecting a larger and more representative dataset from different areas of Rwanda, adding more relevant housing features, and investigating extreme property prices. Other machine-learning algorithms, such as Decision Tree Regression, Random Forest Regression, and Gradient Boosting Regression, could be tested and compared with Multiple Linear Regression. Cross-validation could also be applied to obtain a more reliable assessment of model performance. Finally, the deployed application could be improved by regularly updating the dataset, monitoring prediction accuracy over time, and retraining the model as market conditions change.